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EPSON OPOS ADK MANUAL

TM Storage OCX MANUAL

TM Storage OCX

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Contents

Section 1. Outline

Controls the data that are stored in the Storage Memory of the device.
Loading of OCX enable the application to read and delete the data, also acquire the ID list stored in the Storage Memory of the device.

Section 2. Operating Environment

EPSON OPOS ADK operates under the following environment.

- Computer Hardware

  IBM PC/AT or compatible
- Operating Systems

  Windows 2000

  Windows XP

  Windows Vista
- Accessible Serial Ports

  COM1, COM2, COM3, COM4 (extended port functions allow use of COM5 though COM10)
- Accessible Parallel Ports

  LPT1, LPT2, LPT3 (operation is unverified in LPT3)
- Accessible USB Ports

  USB equipment must be connected to the IBM PC/AT compatible machines that are equipped with USB connectors or use USB expansion cards (however, the operating system must be the English version of Windows 2000, Windows XP or Windows Vista when USB is used.)

  If the USB device cannot be identified when an IBM PC/AT compatible machine equipped with a USB connector is in use contact the manufacturer.
- Accessible Network

  Ethernet support applies to TCP/IP compliant networks.

  The setup of a compliant network should be handled by a qualified network administrator.
- Supported Languages

  Microsoft Visual Basic Version 5.0 or above

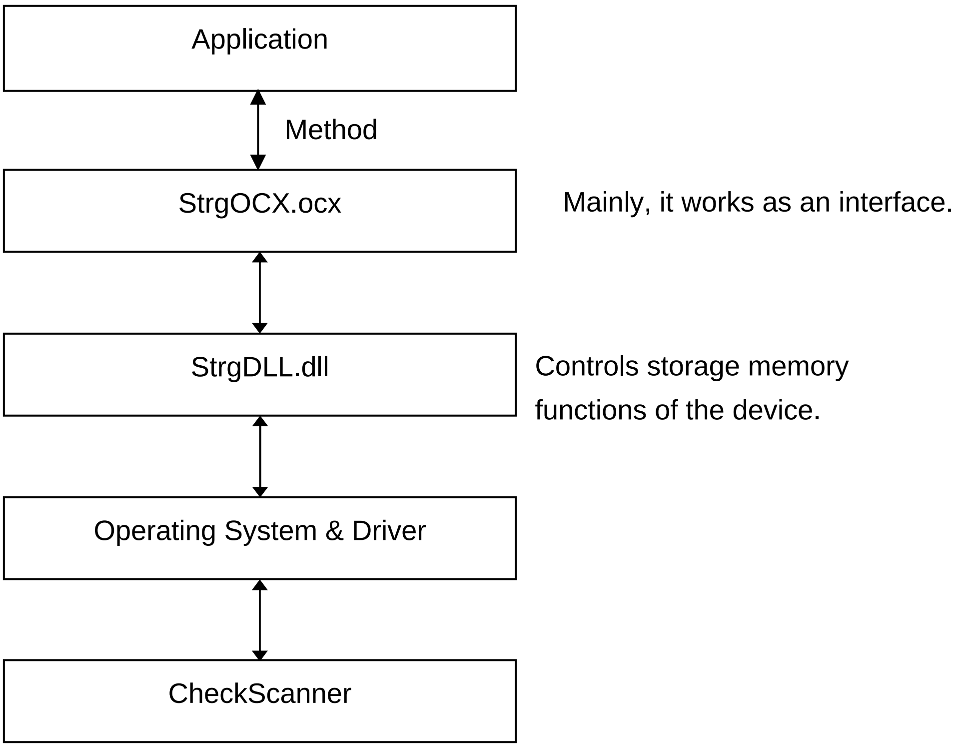
  Microsoft Visual C++ Version 5.0 or above

- Required Files
  - Porthcom.dll
  - Portlpt.dll
  - PortMg11.dll
  - PortNet.dll
  - PortUsb.dll
- Accessible Devices

(Serial)	TM-H6000II with Scanner(256)
	TM-H6000III with Scanner
(Parallel)	TM-H6000IIP with Scanner(256)
	TM-H6000III with Scanner
(USB)	TM-H6000IIU with Scanner(256)
	TM-H6000III with Scanner
(EtherNet)	TM-H6000IIE with Scanner(256)
	TM-H6000III with Scanner

Section 3. Structure

provides an method that controls the Storage Memory of the device to application.
The application requests transactions through Method and the transactions are implemented through layers of systems as follows:



StrgOCX.ocx works as an interface. It handovers transaction task to StrgDLL.dll when Method is called by the application.

StrgDLL.dll functions the management of the Device’s Storage Memory. It communicates with Printers and controls the Storage Memory upon request of transactions.

Section 4. Method

4.1. Method List

The methods provided by are as follows:

See 4.2 for detailed explanation.

Method

Name	**Type**	**Function**
DeleteImage	LONG	Deletes the data stored in the Storage Memory of the device.
RetransmissionImage	LONG	Acquires the data stored in the Storage Memory of the device
GetImageList	LONG	Acquires a list of Image data's IDs that are being stored in the Storage Memory of the device.
PortClose	LONG	Closes a port
PortOpen PortOpen2 PortOpen3	LONG	Opens a port

4.2. Method Description

DeleteImage Method

Syntax **LONG** DeleteImage **(LONG** *DataID***);**

Parameter	Description
	DataID Specifies the *DataID* to be deleted that are stored in the Storage Memory of the device. When the *DataID* is “0”, deletes all the data. When the *DataID* is larger number than “1”, delete the data of specified ID. (Available Range: 1 <= *DataID* <= 65534)

Remarks Delete the specified *DataID* stored in the Storage Memory of the device.

No transaction occurs if no data is available at the specified *DataID.*

In that case, ESTRG_E_NODATA is configured as a return value.

Do not off the power while DeleteImage method is executed.

Behavior of the device is not guaranteed afterward.

If the cover of the device is opened while DeleteImage method is executed, there are cases that the result of the transaction (whether it's succeeded or not) is not confirmed.

The error may returns even if the actual transmission is successful.

Return One of the following is configured.

Value	Meaning
ESTRG_SUCCESS	The Specified data are deleted successfully.
ESTRG_E_BADPARAM	The specified parameter is illegal.
(The specified parameter is outside of the available range)	
ESTRG_E_BUSY	The specified Printer is busy.
ESTRG_E_NODATA	No data at the specified ID.
ESTRG_E_FAILURE	Failed to delete the specified data.
ESTRG_E_NOTPORTOPEN	The specified port is not opened.

RetransmissionImage Method

Syntax **LONG RetransmissionImage (LONG** *DataID*, **BSTR*** *pSaveData,* **DWORD** *Timeout,* **LONG*** *pFormat* **);**

Parametar	Description
DataID	Specifies *DataID* to be loaded from the Storage Memory of the device. (Available Range :1 <= *DataID* <= 65534)
pSaveData	Specifies the address of the memory for setting image data. *Timeout* Sets time-out period between image data request to another image data request by msec unit. If the data are not acquired within the specified period, ESTRG_E_FAILURE is returned. (Available Range: 1 <= *Timeout* <= 65535)
pFormat	The pointer of the loaded data format storage. The data format values to be stored are as follows:

Value	Meaning
STRG_IF_UNKNOWN	Unknown Format
STRG_IF_TIFF	TIFF Format

STRG_IF_JPG	JPEG Format
STRG_IF_BMP	Bitmap Format

Remarks Acquires the data corresponding to the specified *DataID.* An error value is set when the configured value is outside of the available range.

Return One of the following Values is returned.

Value	Meaning
ESTRG_SUCCESS	Succeeds data acquirement
ESTRG_E_BADPARAM+1	Specified *DataID* Parameter value is illegal.
(The specified parameter is outside of the available range)	
ESTRG_E_BADPARAM+3	Specified *Timeo*ut parameter value is illegal.
(The specified parameter is outside of the available range)	
ESTRG_E_BUSY	The specified Printer is busy.
ESTRG_E_NODATA	No data at the specified ID.
ESTRG_E_FAILURE	Failed to acquire the data. Failed to acquire the specified data within time-out period
ESTRG_E_NOTPORTOPEN	The specified port is not opened.

GetImageList Method

Syntax **LONG GetImageList (LONG*** *pListNum,* **BSTR*** *pListData* **);**

Parameter	Description
pListNum	Acquired ID numbers are stored.
pListData	Specifies an memory address for setting ID list of the data The null character is configured when no data is available. Information is set according to the following format. Data ID1(1~65534) , ,Data IDk Ex. When the data are retained at ID1 & 2, stored as "1,2".

Remarks Acquires the total number and the list of the data retained in the Storage Memory of the device.
ESTRG_SUCCESS is returned upon succeeding acquisition of the ID list even though there is no data in the Storage Memory.

Return One of the following Values is returned.

Value	**Meanings**
ESTRG_SUCCESS	Succeeds data acquirement
ESTRG_E_BUSY	The specified Printer is busy.
ESTRG_E_FAILURE	Failed to delete the specified data.
ESTRG_E_NOTPORTOPEN	The specified port is not opened

PortClose Method

Syntax **LONG PortClose();**

Remarks Closing a port

Return One of the following Values is returned.

Value	**Meaning**
ESTRG_SUCCESS	The port is closed successfully.
ESTRG_E_NOTPORTOPEN	The port is not opened.

PortOpen Method

Syntax **LONG PortOpen**(**LONG** *PortType*, **BSTR** *PortName*, **LONG** *BaudRate*, **LONG** *DataBits*, **LONG** *Parity*)

Parameter	Description
PortType	Specifies the port type to be opened. The values of *PortType* parameter are as follows:

Value	**Meaning**
STRG_PORT_COM	Serial
STRG_PORT_LPT	Parallel
STRG_PORT_USB	USB
STRG_PORT_NET	Ethernet

PortName	Specifies the port name to be opened Ex.　To open COM1, specify "COM1".
BaudRate	Configures communication speed. The values of *BaudRate* parameter are as follows:

Value	**Meaning**
STRG_COM_B2400	2400 dps
STRG_COM_B4800	4800 dps
STRG_COM_B9600	9600 dps
STRG_COM_B19200	19200 dps
STRG_COM_B38400	38400 dps
STRG_COM_B56000	56000 dps
STRG_COM_B57600	57600 dps
STRG_COM_B115200	115200 dps

DataBits	Specifies the byte length. The values of *DataBits* parameter are as follows:

Value	**Meaning**
STRG_COM_DATABITS_7	7
STRG_COM_DATABITS_8	8

Parity	Configures the parity bit. The values of *Parity* parameter are as follows:

Value	**Meaning**
STRG_COM_PARITY_NONE	None
STRG_COM_PARITY_EVEN	Even

STRG_COM_PARITY_ODD	Odd

Remarks Opens a port according to the specified *PortName*.
Parameter configurations cannot be omitted even if the *PortType* is specified other than STRG_PORT_COM.
Therefore, All the parameters' configuration is required when STRG_PORT_COM is specified as *PortOpen method*.

Return One of the following Values is returned.

Value	Meaning
ESTRG_SUCCESS	The specified port is opened successfully.
ESTRG_E_BADPARAM+1	The specified *PortType* value is illegal.
ESTRG_E_BADPARAM+3	The specified *BaudRate* value is illegal.
ESTRG_E_BADPARAM+4	The specified *DataBits* value is illegal.
ESTRG_E_BADPARAM+5	The specified Parity value is illegal.
ESTRG_E_BADPORT	The specified Device Port is illegal.
ESTRG_E_PORTUSED	The port is being used.
ESTRG_E_REOPEN	The port is already open.

PortOpen2 Method (Hidden Method)

Syntax **LONG PortOpen(LONG** *PortType*, **BSTR** *PortName*,**);**

Parameter	Description
PortType	Specifies the port type to be opened. The values of *PortType* parameter are as follows:

Value	Meaning
STRG_PORT_LPT	Parallel
STRG_PORT_USB	USB
STRG_PORT_NET	Ethernet

PortName	Specifies the port name to be opened Ex. To open USB1, specify “USB1”.

Remarks Opens a port according to the specified *PortName*.
Applied for using Parallel, USB, and Ethernet Device.
(Unable to specify according to the serial device.)

Return One of the following Values is returned.

Value	Meaning
ESTRG_SUCCESS	The port is opened successfully.
ESTRG_E_BADPARAM+1	The specified *PortType* value is illegal.
ESTRG_E_BADPORT	The specified device port is illegal.
ESTRG_E_PORTUSED	The port is being used.
ESTRG_E_REOPEN	The port is already open.

PortOpen3 Method (Hidden Method)

Syntax **LONG PortOpen(BSTR** *DeviceName***);**

Parameter	Description
DeviceName	Specifies the device or logical device to open.

Remarks Opens the port corresponding the specified device.

Return One of the following Values is returned.

Value	Meaning
ESTRG_SUCCESS	The port is opened successfully.
ESTRG_E_BADPORT	The specified device port is illegal.
ESTRG_E_PORTUSED	The port is being used.
ESTRG_E_REOPEN	The port is already open.

Section 5. Programming

The following are the examples of TM Storage OCX Programming. Error transactions are omitted.

1. The acquisition of image data that are restored in the Storage Memory of the device.
   The following are the examples of Cordings for acquiring image data by the RetransmissionImage Method.

```
<< Visual Basic >>
Dim DataID As Long          'Image data's ID
Dim strdata As String       ' Variables to restore image data (Character String
                              Type)
Dim binData() As Byte       ' Variables to restore image data (Byte Type)
Dim lFormat As Long         ' Format for acquired image data
Dim lSize As Long           ' Size of acquired image data
Dim lLoop As Long           ' Variables for loop
Dim lRet As Long            ' Return Value

    ' Acquiring image data by the RetransmissionImage method
        lRet = StrgOCX1.RetransmissionImage(DataID, strdata, 2000, lFormat)
    ' Acquires the length of binary data
        lSize = LenB(strdata)
    ' Secure the memory space for the acquired data
        ReDim binData(lSize)
    ' Sets data at the allocated memory
        For lLoop = 0 To lSize - 1
            binData(lLoop) = AscB(MidB(strdata, (lLoop + 1), 1))
        Next lLoop
```

<< Visual C++ >>

```
BSTR      bstData;     // Variables to restore image data
BYTE*     byData;      // Pointer of image data
long      lRet;        // Return value
long      lSize;       // Size of image data
long      lFormat;     // Format of image data

// Acquires data by RetransmissionImage method RetransmissionImage
lRet = StrgOCX1.RetransmissionImage(DataID, &bstData, 2000, &lFormat)
// Acquires the length of binary data
lSize = ::SysStringByteLen(bstData);
// Secures the memory space for the acquired data
byData = new BYTE[lSize];
// Sets data at the allocated memory
memcpy(byData, bstData, lLen);
//
.....
//
// Frees the allocated memory
::SysFreeString(bstData);
delete[] byData;
```

Section 6. Warnings

Memory Error might occur when it is failed to delete memories by the DeletImage method.
If the error occurs, reboots the device before the retrial.